

Beyond Metaphor:

A Blinded Confirmatory Design and Computational Discovery Engine for Quantum-Structural Analogues in Classical Sanskrit Thought (RISHI-Q)

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RISHI-Q Project

<https://github.com/arjunkshah/rishi-q>

Sister experimental project: RISHI-FIELDS (<https://github.com/arjunkshah/rishi-fields>)

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Working methods + discovery manuscript. Confirmatory H1 has not been unlocked. Human validation and OSF preregistration remain outstanding. System B discovery exhibits are exploratory under a heuristic annotator on public-domain development text. Success is not defined as a positive quantum result.

Abstract

Claims that classical Sanskrit philosophical texts “anticipated quantum mechanics” are culturally pervasive but scientifically fragile: once modern physics is known, ambiguous metaphysical language can be reinterpreted until it sounds quantum-like. RISHI-Q (Research Instrument for Structural Historical Inference of Quantum-analogues) is a dual-system computational research project. System A is a preregisterable confirmatory experiment asking whether selected Sanskrit corpora exhibit quantum-specific structural correspondence with QM/QFT beyond matched historical controls, after blinding, translation controls, vocabulary masking, evidence-constrained annotation, and cluster-aware inference. System B is a computational discovery engine that extracts evidence-bound concept graphs, mines unsupervised Rishi Motifs without physics labels, then maps motifs to physical theories only post hoc—searching for previously unquantified structural patterns, translation effects, anomalies, and claim–data divergences. The framework separates Level I (generic metaphysical), Level II (classical/field-like), and Level III (quantum-specific) similarity and forbids upgrading metaphors such as unity into entanglement. This manuscript documents the scientific design, ontology, pipelines, reproducibility machinery, instrument validation on synthetic and modern-physics controls, an exploratory public-domain pilot ($n = 276$), a Capra-style claim autopsy (5/5 claims CONTRADICTED_AS_QUANTUM on the Vedānta PD slice), and System B discovery outputs under a heuristic annotator. It does not claim confirmatory Sanskrit results, historical discovery of quantum mechanics, or Tier-5 transformative discoveries. A negative confirmatory result remains a successful scientific outcome. Physical detection of prāṇa or related concepts is out of scope and is pursued separately in RISHI-FIELDS.

Keywords: computational humanities; history of physics; Sanskrit philosophy; quantum foundations; blinded annotation; motif discovery; falsifiability

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1 Introduction

1.1 Motivation

Popular and semi-scholarly literature frequently maps Sanskrit concepts—prāṇa, ākāśa, śakti, Brahman, spanda, aṇu/paramāṇu—onto energy, fields, vibration, or quantum phenomena Capra [1975]. Such comparisons are historically interesting as cultural reception, but they are poor evidence of structural correspondence unless they survive controls that make retrospective reinterpretation difficult Restivo [1983].

1.2 Two complementary systems

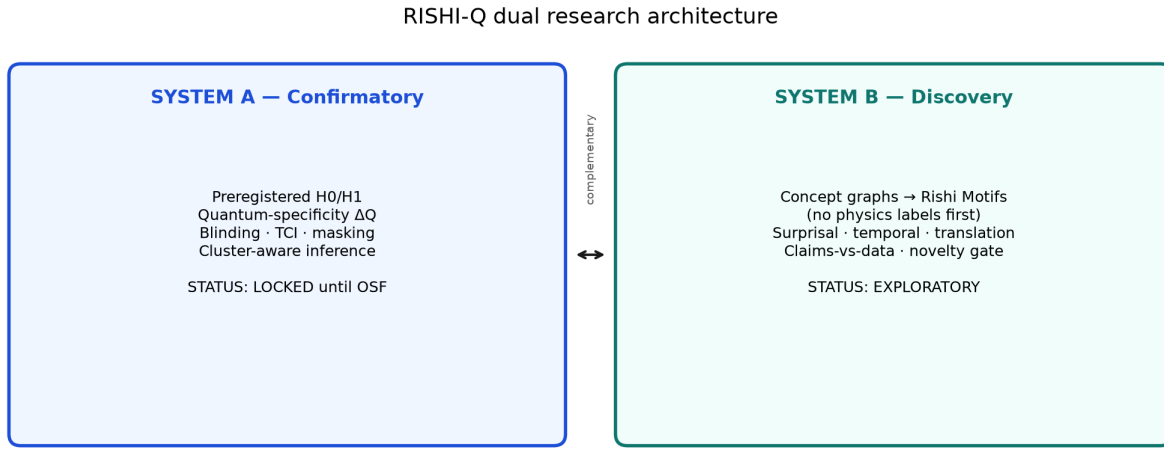


Figure 1: RISHI-Q dual architecture: confirmatory hypothesis testing (System A) and unsupervised structural discovery (System B).

System A preserves a sharp confirmatory question. System B ensures the project is not limited to confirming or rejecting an already-discussed claim: it searches for structural motifs and historical patterns that were not pre-specified as “quantum.”

1.3 Research questions

Confirmatory (System A). Do selected classical Sanskrit philosophical corpora exhibit statistically greater structural correspondence with modern QM/QFT than matched historical philosophical controls?

Discovery (System B). Which recurring structural motifs, temporal combinations, translation shifts, anomalies, or claim–data divergences are quantitatively supported in the sampled corpora— independent of physics labels at discovery time?

1.4 What RISHI-Q is not

RISHI-Q is not a proof that scriptures discovered quantum mechanics; not a cosine-similarity demo between Upaniṣads and physics encyclopedias; and not a spiritual affirmation project. Even

strong Level III enrichment would establish at most a controlled structural analogy, not historical discovery.

1.5 Design principle

The experiment must be capable of producing a convincing negative result. Methodological choices made only to make Sanskrit look more quantum-like are forbidden (protocol/decisions.md). Success is not defined as a positive quantum result; it is defined by rigorous search and honest reporting, including original negative or claim-contradicting findings.

2 Related Work

2.1 Philosophy of quantum theory

Level III definitions follow contemporary philosophy of QM/QFT rather than popular slogans Lewis [2016], Maudlin [2019], Wallace [2012], Ruetsche [2011], Weinberg [1995].

2.2 Classical Indian ontologies

Atomism, substance, transformation, and substrate language in Vaiśeṣika, Sāṃkhya, and related schools must be read historically Halbfass [1992], Potter [1977], Larson and Bhattacharya [1987].

2.3 Physics–mysticism critique

Uncontrolled parallelisms are a known sociological pattern Restivo [1983], Capra [1975]. Capra-style analogy is treated as a methodological anti-pattern.

2.4 Computational annotation

LLM annotation can shift label distributions; human audit is mandatory (status: REQUIRES_EXTERNAL_HUM

3 Conceptual Framework

3.1 Three levels of similarity

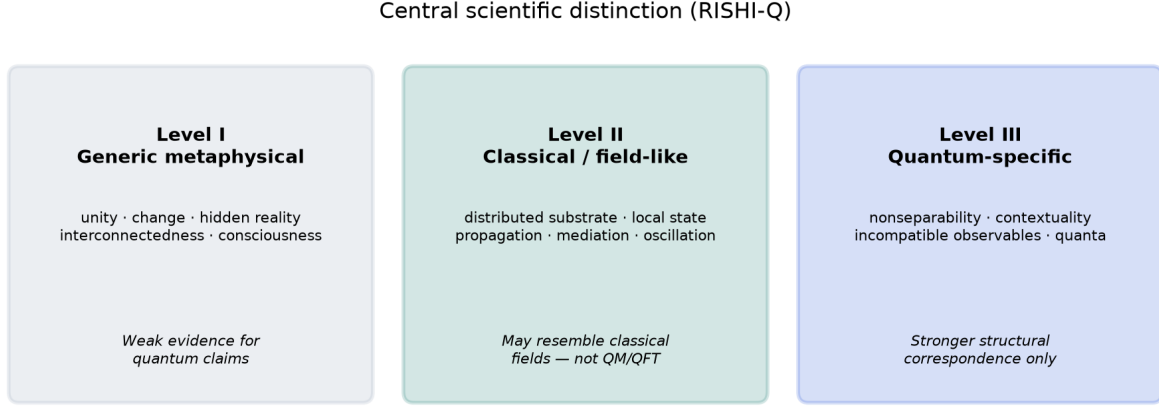


Figure 2: Central scientific distinction: Level I vs Level II vs Level III.

- Level I — Generic metaphysical: unity, change, hidden reality, interconnectedness. Weak for quantum claims.
- Level II — Classical / field-like: distributed substrate, local state, propagation, oscillation, mediation. May resemble classical fields Jackson [1999], not automatically quantum.
- Level III — Quantum-specific: discrete allowed states, fundamental probability, nonseparability, contextuality, incompatible observables, measurement-conditioned outcomes, quantized excitations.

3.2 Hard interpretation rules

Unity $\neq$ entanglement; vibration $\neq$ QFT; prāṇa $\neq$ energy; ākāśa $\neq$ quantum field; aṇu $\neq$ elementary particle; observer importance $\neq$ quantum measurement. When ambiguous, prefer NA/U over 1.

Table 1: Hard interpretation rules (annotation codebook).

rule_id	rule
R01	Everything is one does NOT imply entanglement (Q06).
R02	Interconnected does NOT imply nonlocality.
R03	Vibration does NOT imply QFT.
R04	Prana does NOT automatically mean physical energy.
R05	Akasa does NOT automatically mean a physical/quantum field.
R06	Sakti does NOT automatically mean physics energy.
R07	Anu does NOT automatically mean elementary particle.
R08	Observer importance does NOT imply quantum measurement.
R09	Epistemic uncertainty does NOT imply quantum indeterminacy.
R10	Invisible does NOT imply quantum.
R11	When ambiguous, prefer NA/U over 1.
R12	Positive labels require exact evidence spans.

3.3 Discovery success tiers

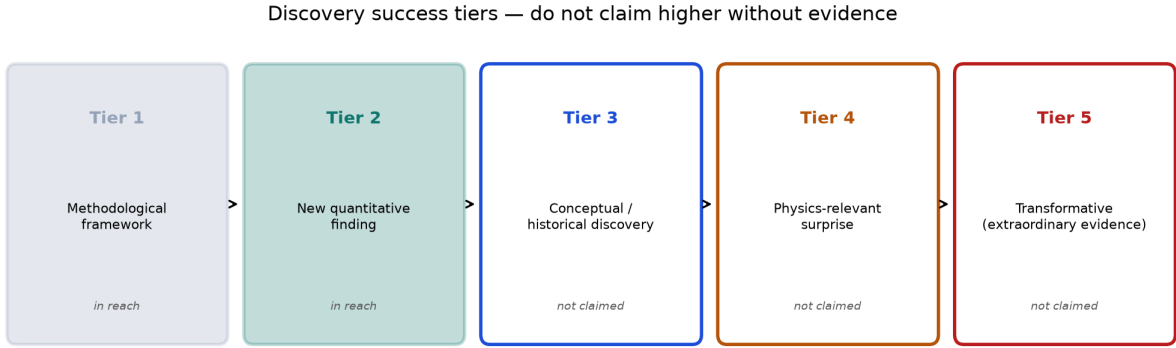


Figure 3: Tiered discovery success. Higher tiers require stronger evidence; Tier 5 is extraordinary.

4 Hypotheses (System A) and Discovery Goals (System B)

Primary confirmatory hypotheses:

$$H_0 : \Delta_Q \leq 0, \quad H_1 : \Delta_Q > 0,$$

$$QS_i = S(X_i, Q) - \max_{C \in \text{Classical}} S(X_i, C), \quad \Delta_Q = \mathbb{E}[QS_{\text{target}}] - \mathbb{E}[QS_{\text{control}}].$$

Secondary hypotheses H2–H8 cover specificity and robustness (protocol/hypotheses.md). Discovery goals D1–D8 cover motif mining, enrichment, surprisal, temporal analysis, translation shifts, claims-vs-data, atlas construction, and novelty gating—explicitly exploratory until frozen and replicated.

Table 2: Project status gates (honest incomplete items listed).

gate	status
Ontology v0.1	PILOTED
Synthetic E2E pipeline	COMPLETE
Positive-control instrument tests	PASSING
Human validation	REQUIRES_EXTERNAL_HUMAN_VALIDATION
OSF preregistration	READY_FOR_EXTERNAL_PREREGISTRATION
Confirmatory analysis	LOCKED
Licensed development corpus (500-800)	IN_PROGRESS_POINTERS
Expert Sanskrit review	REQUIRES_EXPERT_REVIEW
Expert physicist review	REQUIRES_EXPERT_REVIEW

5 Corpora and Firewall

Targets: Upaniṣadic/Vedāntic and related schools. Controls: Buddhist, Greek, Chinese, and other historical philosophical slices; literary negatives; modern physics references for instrument checks.

Development vs confirmatory firewall

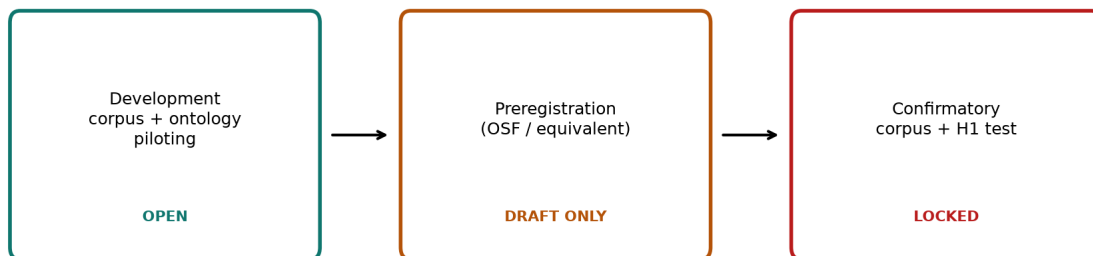


Figure 4: Development vs confirmatory firewall; confirmatory analysis remains locked.

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6 Ontology and Annotation

Ontology v0.1 encodes 36 physics-derived structural features.

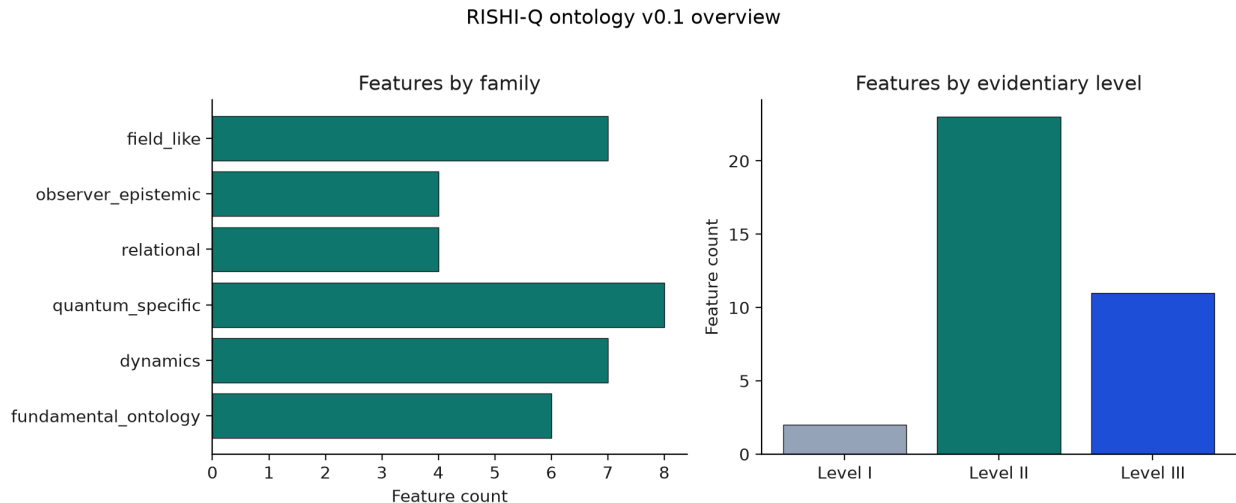


Figure 5: Ontology inventory by family and evidentiary level.

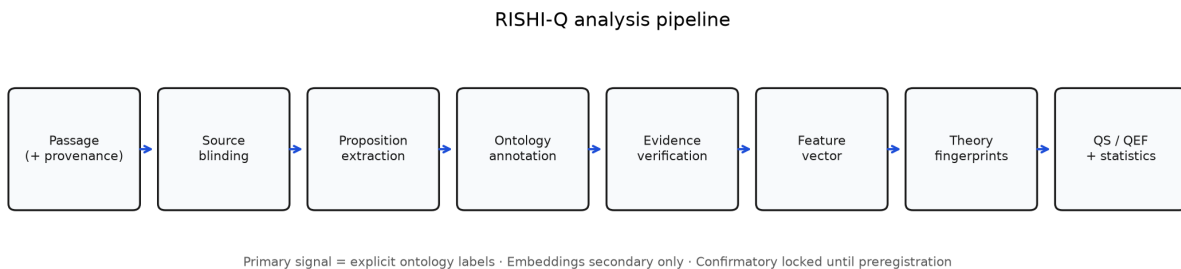


Figure 6: System A analysis pipeline. Embeddings are secondary only.

Positive labels require exact evidence spans. Local development uses a deterministic heuristic backend; Kaggle notebooks support open-model replication.

7 System B: Computational Discovery Engine

7.1 Concept graphs

Passages are represented not only as feature vectors but as evidence-bound concept graphs (nodes such as substrate, state, observer; edges such as pervades, manifests_as). Edges are never invented without positive labels and evidence spans.

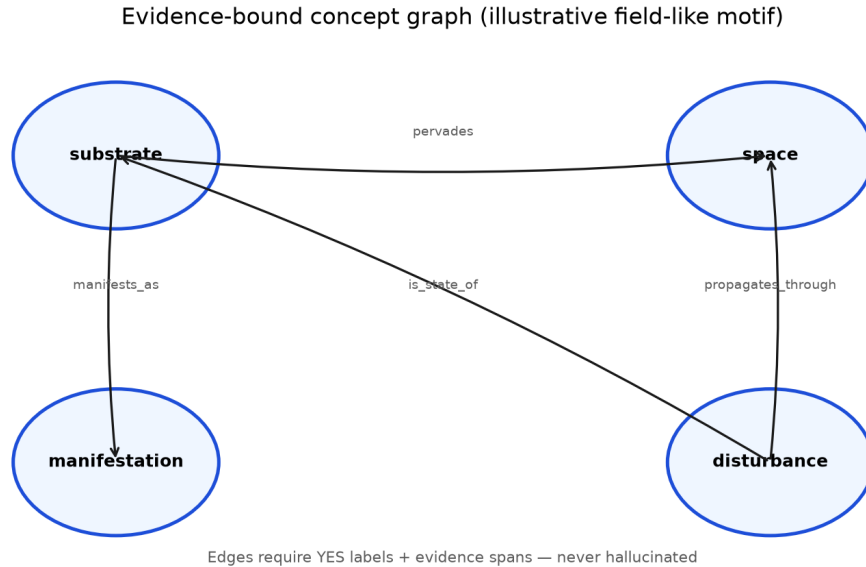


Figure 7: Illustrative evidence-bound concept graph (field-like motif shape).

7.2 Label-free motif discovery

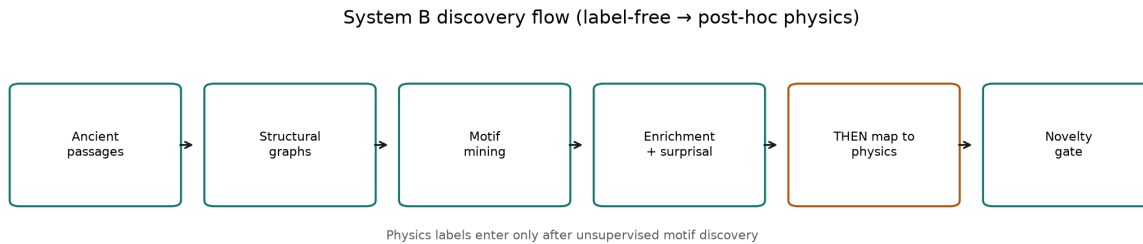


Figure 8: Discovery flow: physics labels enter only after unsupervised motif mining.

Recurring structural signatures are mined as Rishi Motifs. Only afterward are motifs compared to Newtonian, classical-field, thermodynamic, relativistic, QM, and QFT fingerprints.

7.3 Significance without p-hacking

Motif enrichment uses tradition contrasts and work-cluster bootstrap. Ranking combines effect size, replication across works, specificity, and robustness—not p -value primacy. Discovery/replication splits are work-stratified to reduce leakage. Exploratory candidates remain exploratory until frozen and tested on held-out works.

8 Statistics

Inference respects passage $\subset$ section $\subset$ work $\subset$ tradition. Planned tools: mixed-effects models, cluster bootstrap CIs, cluster-preserving permutation tests, BH-FDR for secondary tests, and

power simulation.

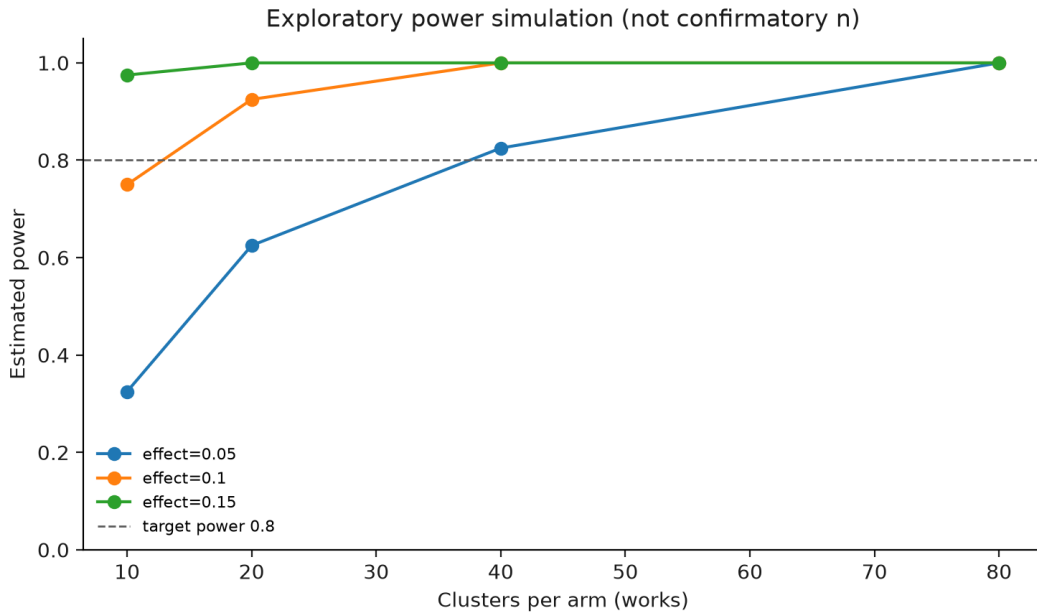


Figure 9: Exploratory power curves. Confirmatory n not frozen.

9 Instrument Validation (Synthetic Controls)

These results validate the measurement instrument, not H1.

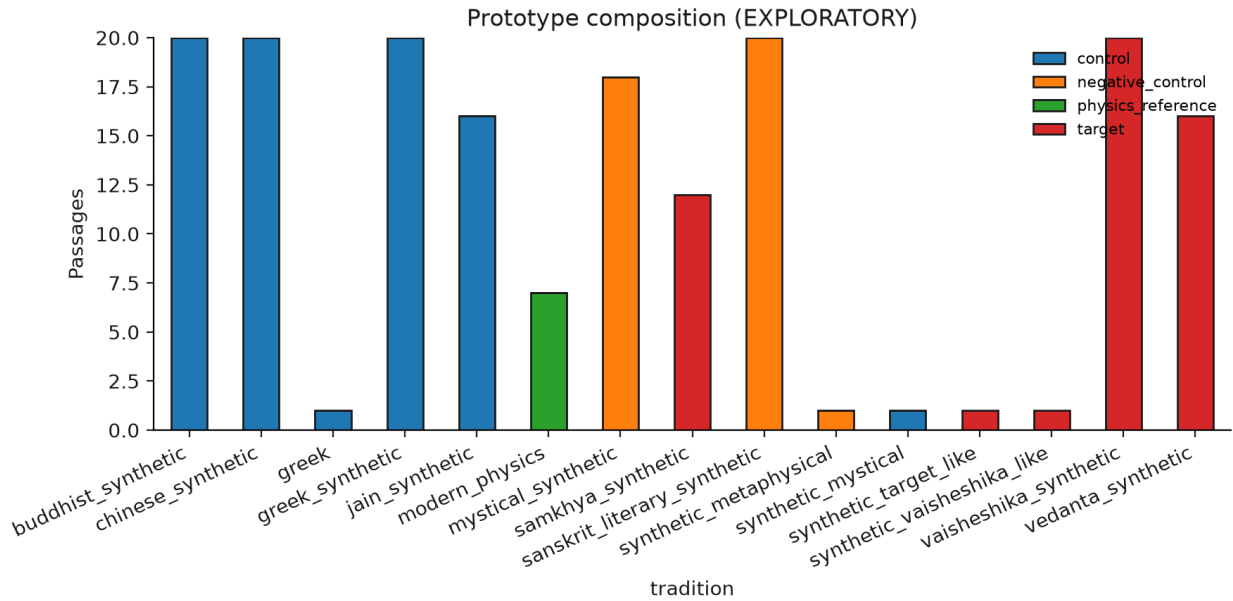


Figure 10: Development prototype composition (protocol §95-style panel; synthetic).

Table 3: Positive-control instrument check on modern physics passages.

passage_id	work	QS	QEF	quantum_mechanics	classical_em
PHYS_NEWTON_001	Synthetic Newtonian description	-0.500	0.000	0.333	0.667
PHYS_EM_001	Synthetic classical EM description	-0.100	0.000	0.000	1.000
PHYS_THERMO_001	Synthetic thermodynamics description	-1.000	0.000	0.000	0.500
PHYS_QM_001	Synthetic QM description	0.900	1.000	1.000	0.000
PHYS_QFT_001	Synthetic QFT description	0.333	1.000	0.200	0.667
PHYS_ENTANGLE_001	Synthetic entanglement description	0.800	1.000	0.800	0.000

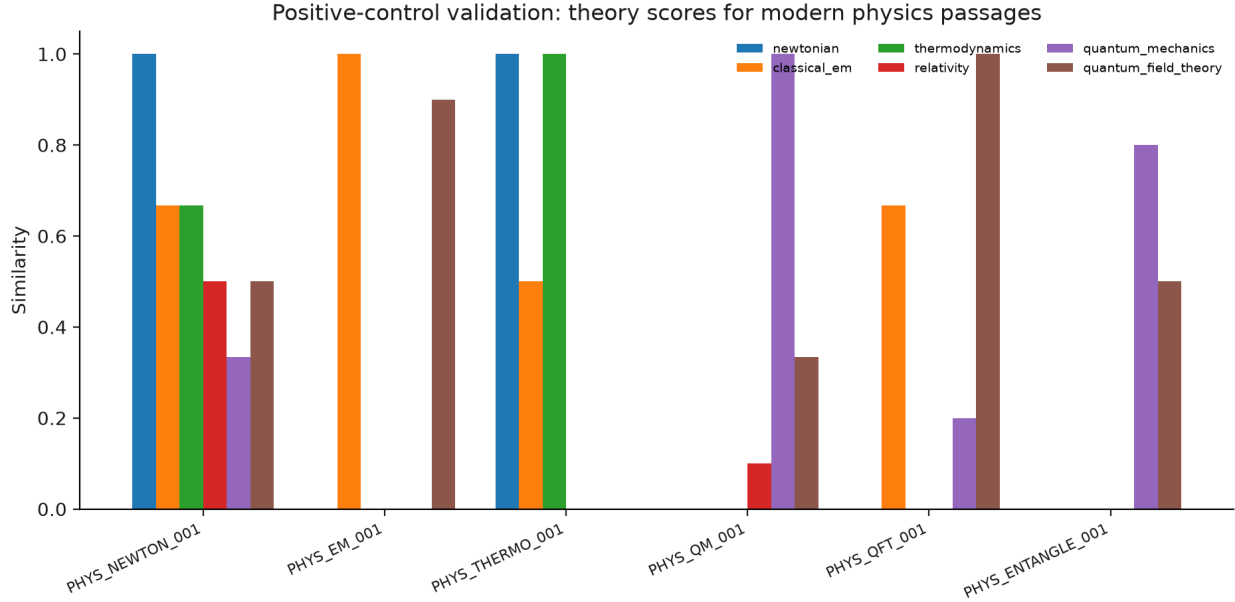
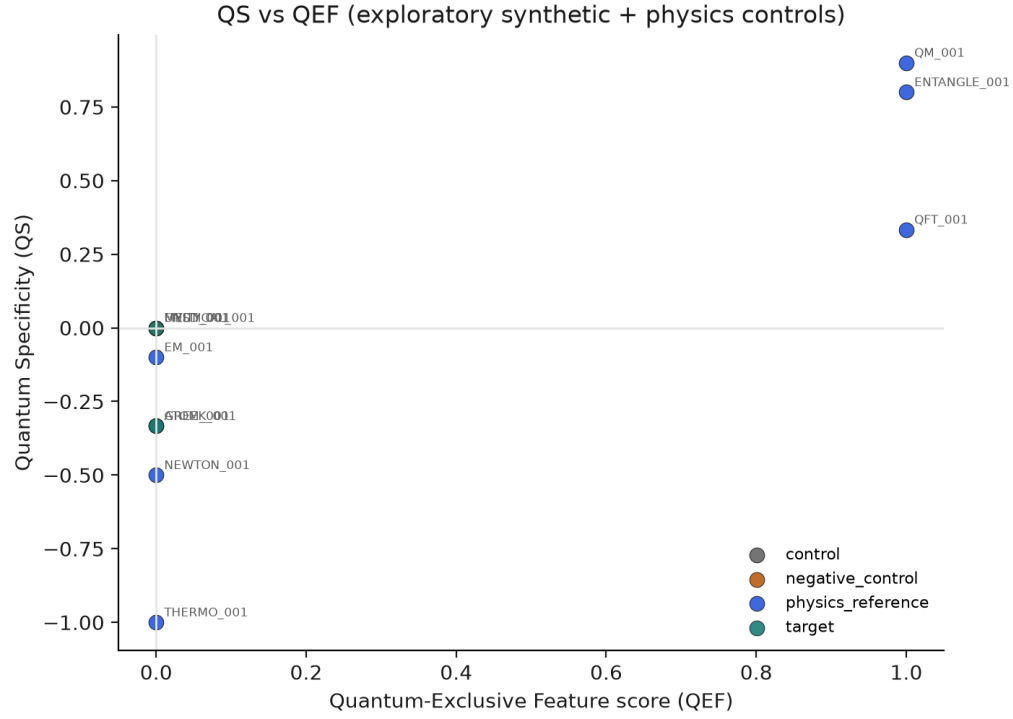
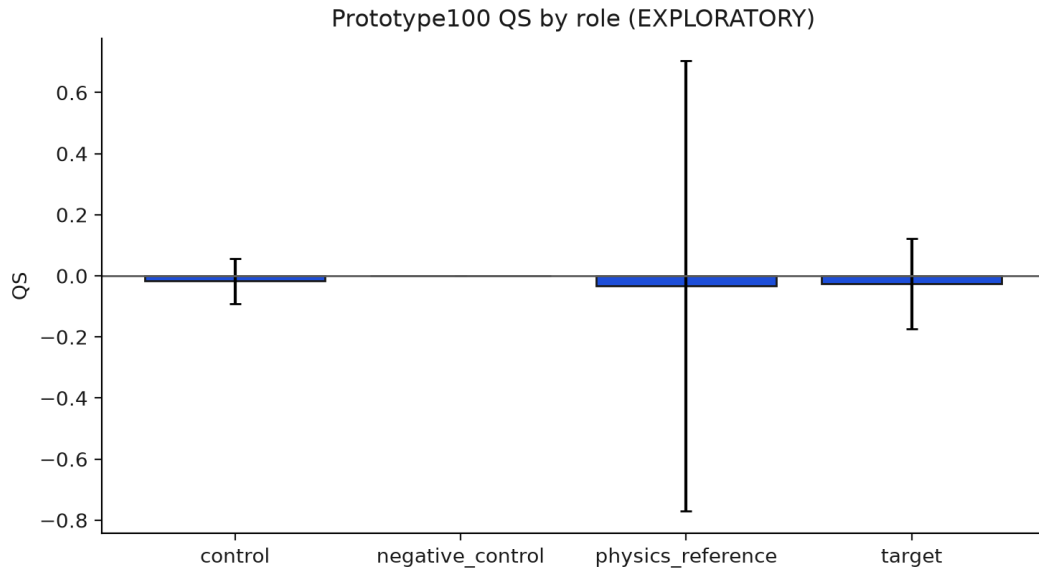


Figure 11: Theory similarities for modern-physics control passages.

Unity metaphors do not trigger nonseparability; QM/QFT controls yield high QS/QEF as required.

Figure 12: QS vs QEF on seed physics/control passages.

On prototype100, target-like synthetic slices are near-null relative to controls ($\Delta_Q \approx -0.015$, $p \approx 0.58$).

Figure 13: QS by role on prototype100 (exploratory).

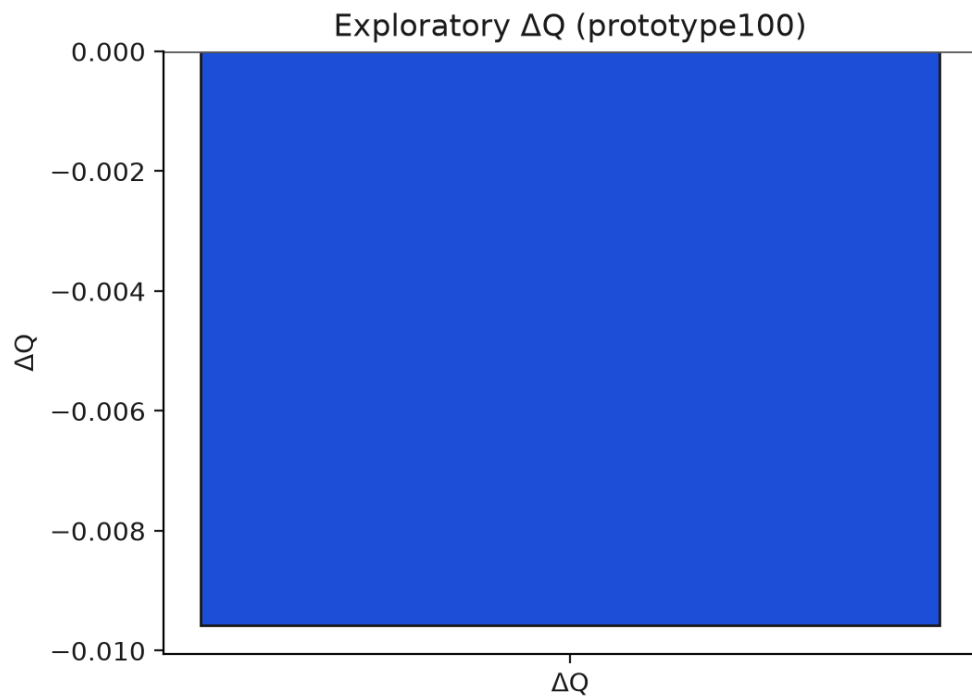
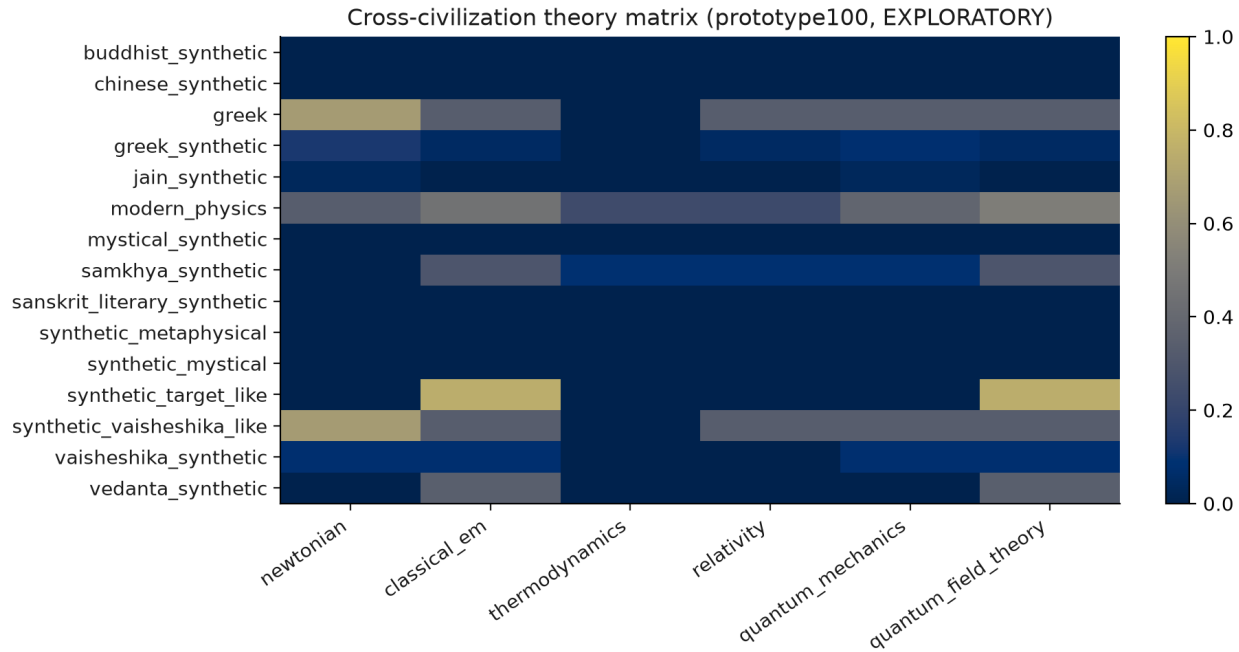
Figure 14: Exploratory ΔQ (not confirmatory).

Figure 15: Cross-civilization theory tournament (exploratory synthetic panel).



Figure 16: Field-ontology class fractions by role.

10 Exploratory Public-Domain Pilot

A public-domain English development panel ($n = 276$: Upaniṣads/Paramananda; Lucretius; Timaeus; Dhammapada; Tao Te Ching; plus physics controls) was annotated with the heuristic backend. Under this annotator, historical slices sit near the annotation floor ($\Delta_Q \approx 0$, $QEF = 0$), while physics controls still discriminate—an instrument limitation finding, not a confirmatory Sanskrit null.

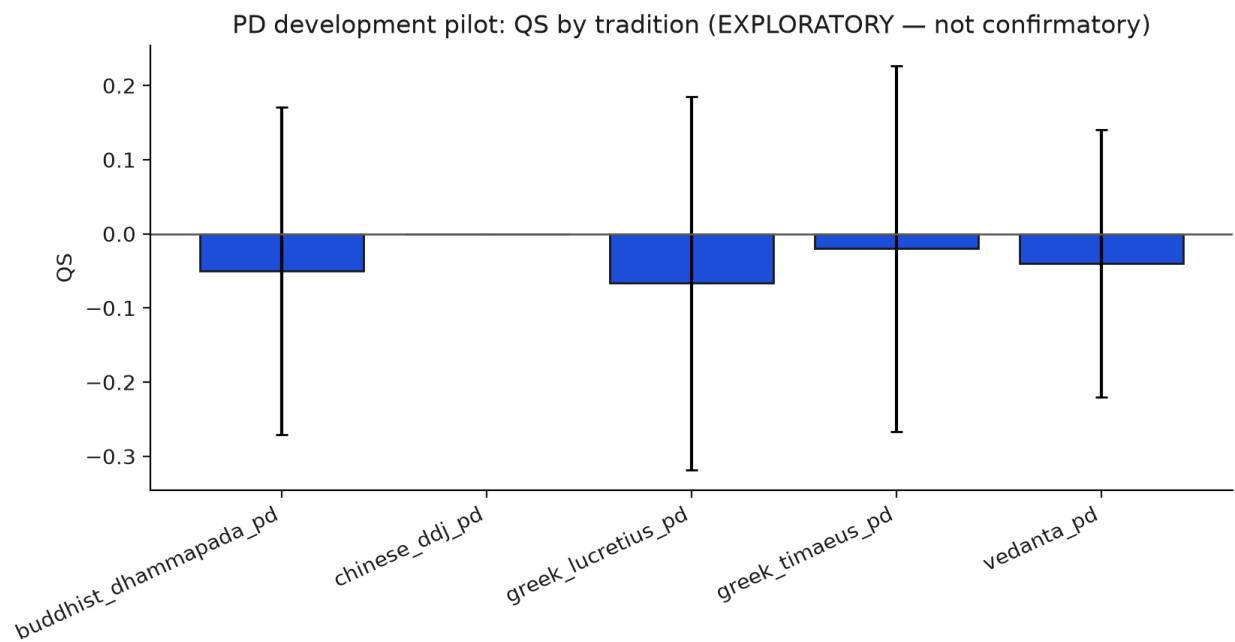


Figure 17: PD pilot QS by tradition (EXPLORATORY).

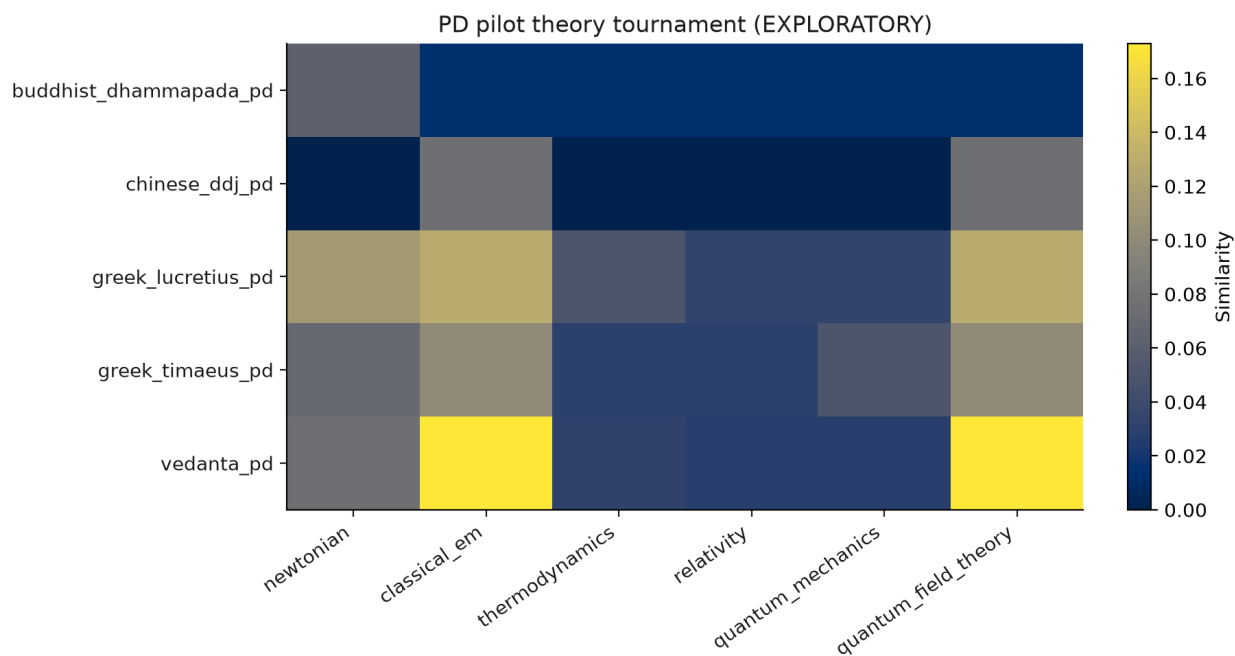


Figure 18: PD pilot theory tournament (EXPLORATORY).

Settlement of H1 requires LLM/human annotation and preregistration.

11 Flagship finding: Capra-style claim autopsy

Popular culture asserts that Sanskrit wisdom anticipated quantum physics. On the public-domain Vedānta panel, a curated autopsy of Capra-style claims finds the reverse pattern:

Five of five curated Capra-style claims are classified **CONTRADICTED_AS_QUANTUM**: Level I/II metaphysical components appear in the sample, while quantum-specific (Level III) components do not. Masking modern physics lexicon does not manufacture quantum signal (mean $\Delta QS \approx +0.006$ on Vedānta). The PD “electrons” sentence is Paramananda’s 1919 commentary rejecting material science as a path to Brahman—the rhetorical opposite of Capra-style co-option.

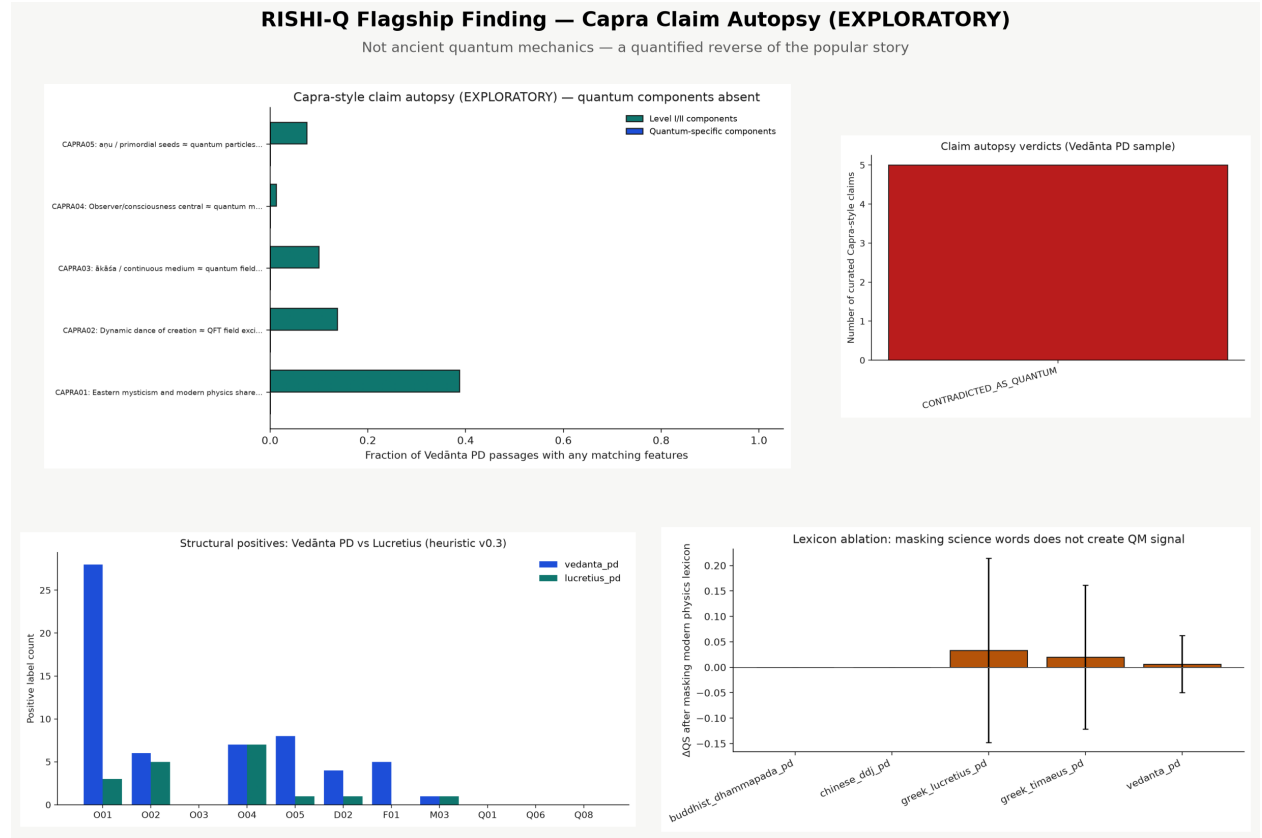


Figure 19: Flagship exploratory poster: Capra autopsy, tradition contrast, lexicon ablation.

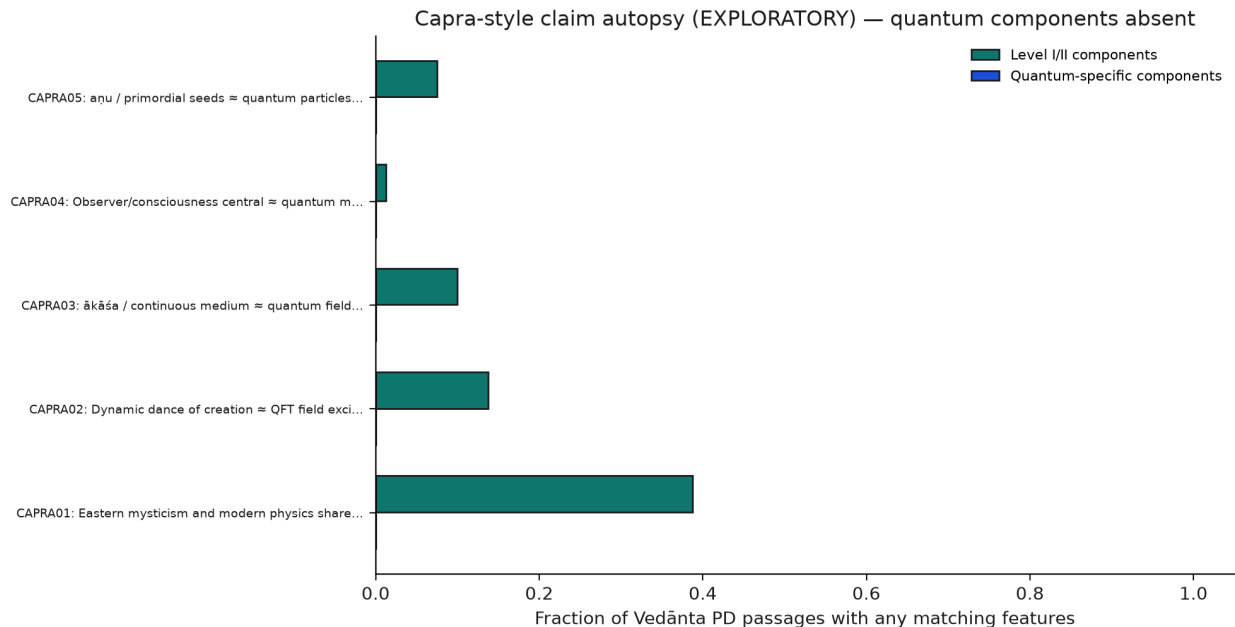


Figure 20: Capra-style claim autopsy on Vedānta PD (EXPLORATORY).

This is a Tier 2–3 quantitative finding about how quantum–Sanskrit rhetoric works—not a Tier 5 claim that ancient authors discovered QM. Open-model annotation on Kaggle was subsequently authenticated under account aks1321; early GPU batch workers did not attach accelerators, and a CPU oneshot pilot returned uniformly NA labels (model too weak for 36-feature oneshot JSON). A batched CPU follow-up and GPU unlock (phone verification / UI accelerator) remain part of the exploratory annotation path. The confirmatory firewall is unchanged.

12 Flagship exploratory finding

We did not find that the Upaniṣads contain quantum mechanics. Under heuristic annotator v0.3 (classical natural-philosophy cues + Level I metaphysical cues; no quantum-family shortcuts), three results cohere:

1. Vedānta PD English shows substantial Level I substrate/manifestation structure (Brahman/Ātman language) with zero historical Q-family positives in this sample, while modern-physics controls retain Q-family hits.
2. Lucretius shows classical void/composition positives under the same annotator—detectable Level I/II natural philosophy without auto-promotion to quantum.
3. The Paramananda PD sample contains modern scientific anachronisms (e.g. “electrons”) in commentary language: apparent scientific resonance can be editorial modernization.

Target–control Δ_Q remains near null.

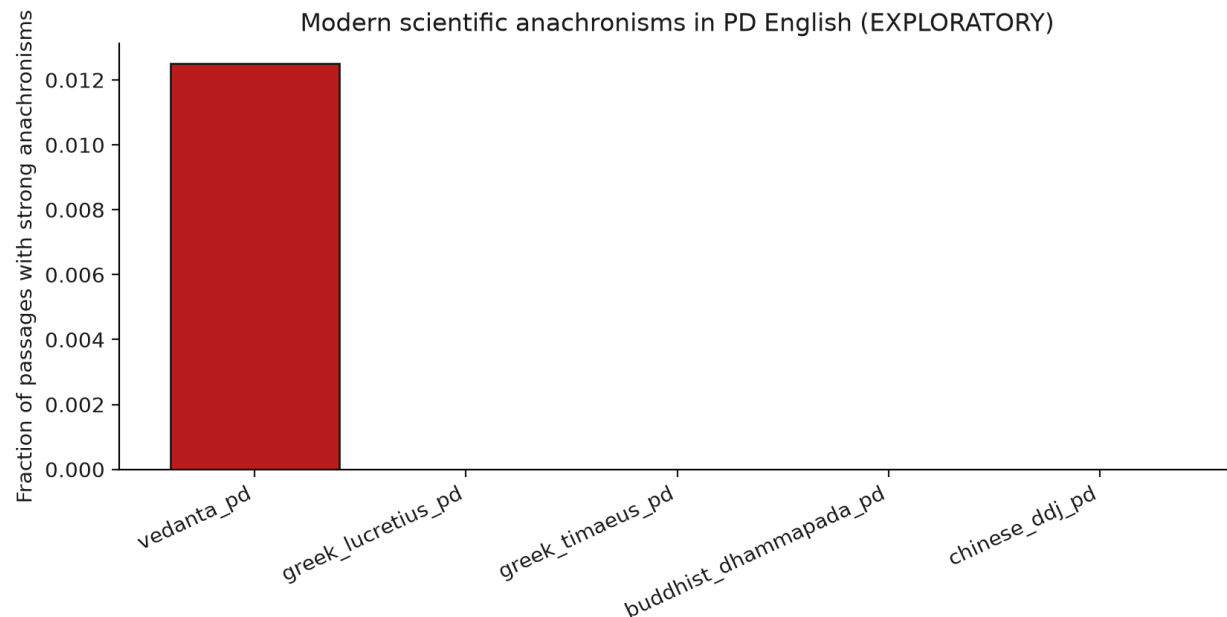


Figure 21: Strong modern-science anachronisms by tradition (EXPLORATORY).

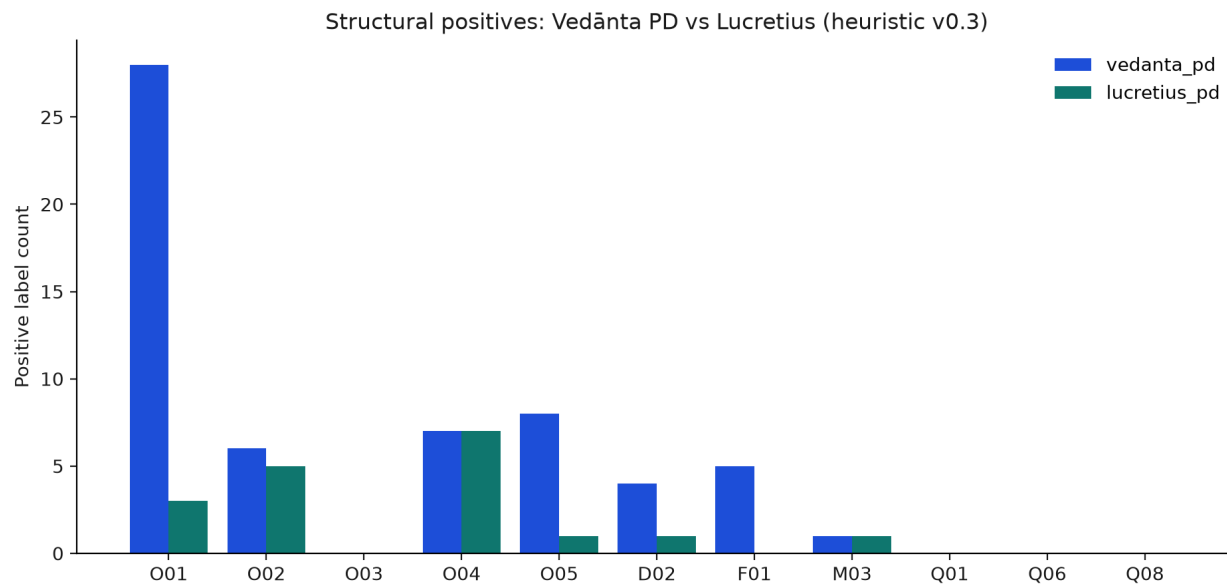


Figure 22: Structural feature positives: Vedānta PD vs Lucretius (heuristic v0.3).

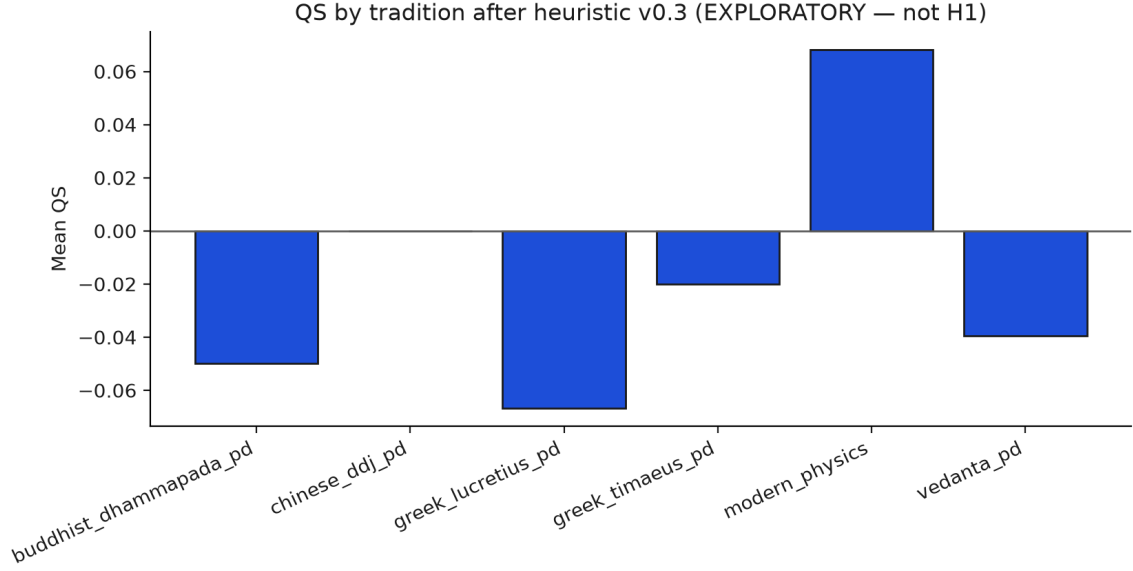


Figure 23: QS by tradition after heuristic v0.3 (EXPLORATORY — not confirmatory H1).

This triad is a Tier 2 quantitative finding about how quantum–Sanskrit rhetoric mixes meta-physics, classical analogies, and modern editorial vocabulary—not a Tier 5 ancient-discovery claim.

13 Exploratory System B Findings

13.1 Motifs and enrichment

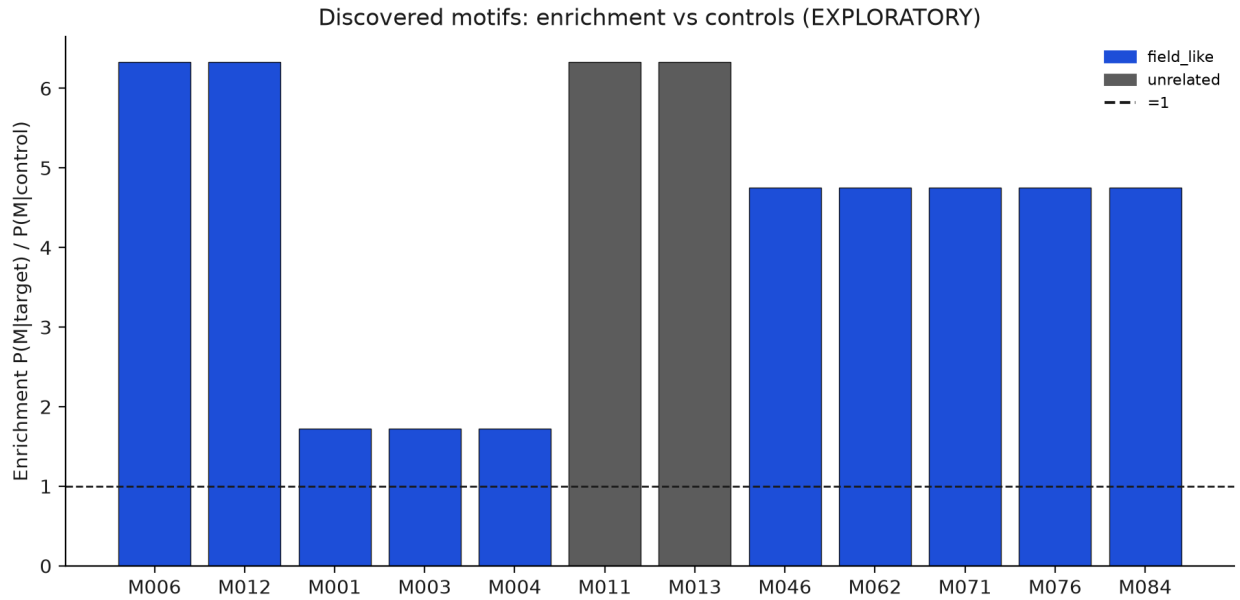


Figure 24: Top discovered motifs with post-hoc physics families (EXPLORATORY).

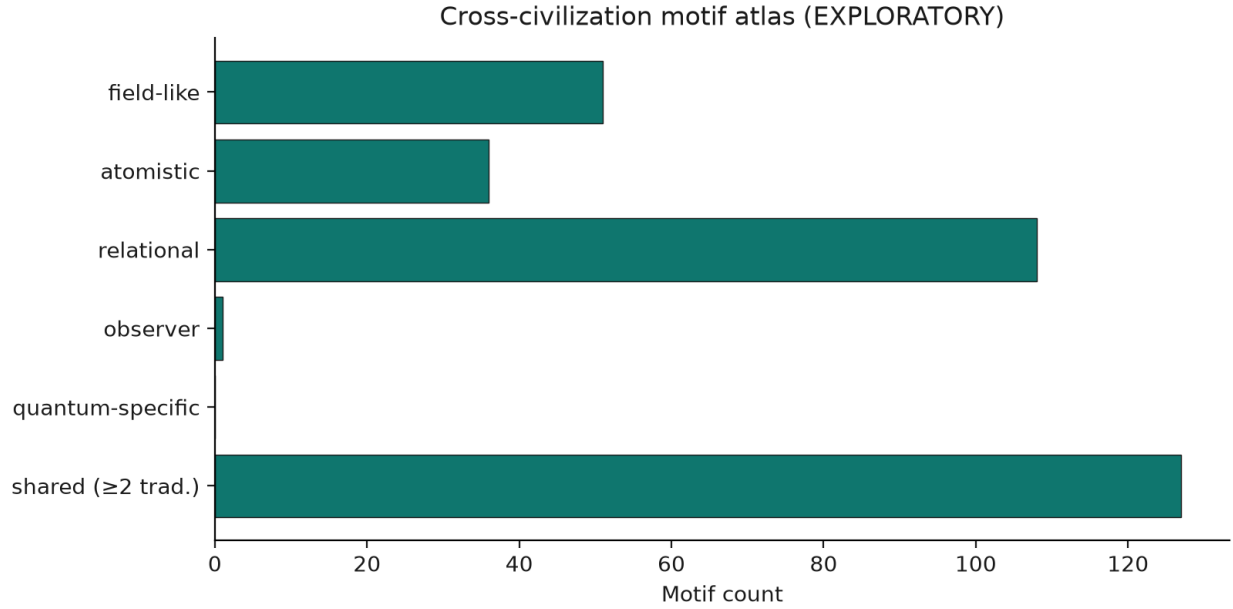


Figure 25: Cross-civilization motif atlas categories.

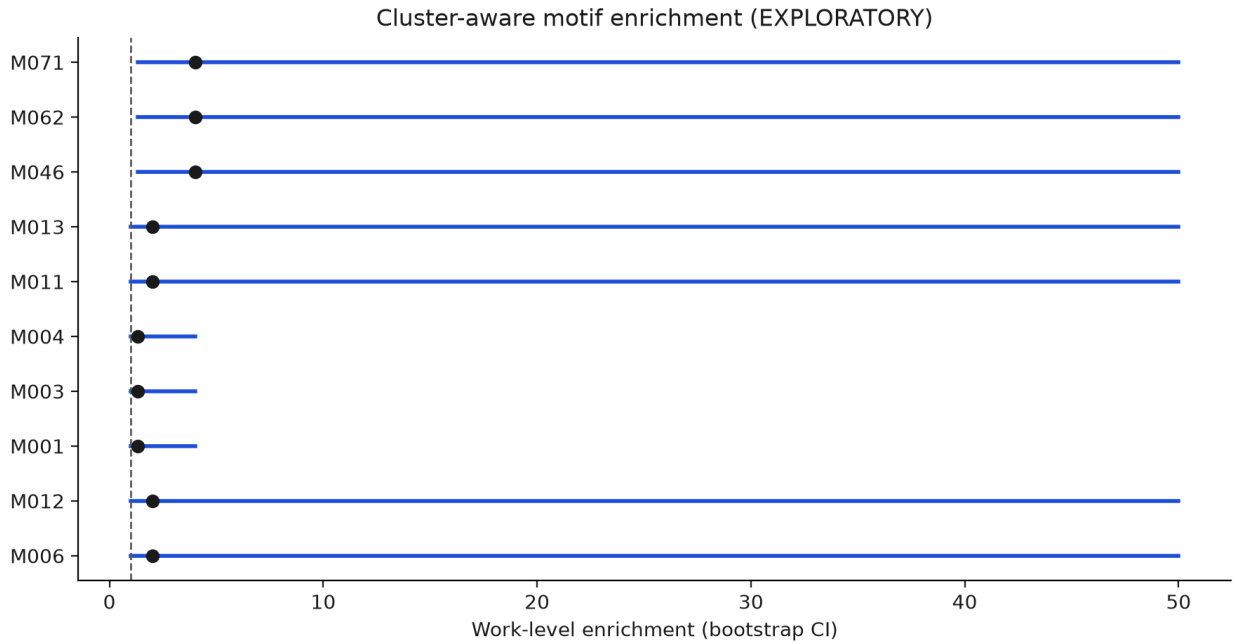


Figure 26: Work-cluster bootstrap enrichment intervals for top motifs.

Under the heuristic PD annotations, no quantum-specific motifs were identified. Several field-like motifs appear preferentially in historical controls rather than the Sanskrit target slice—a claim-weakening pattern for Sanskrit-specific quantum readings of those structures.

13.2 Surprisal, temporal, translation

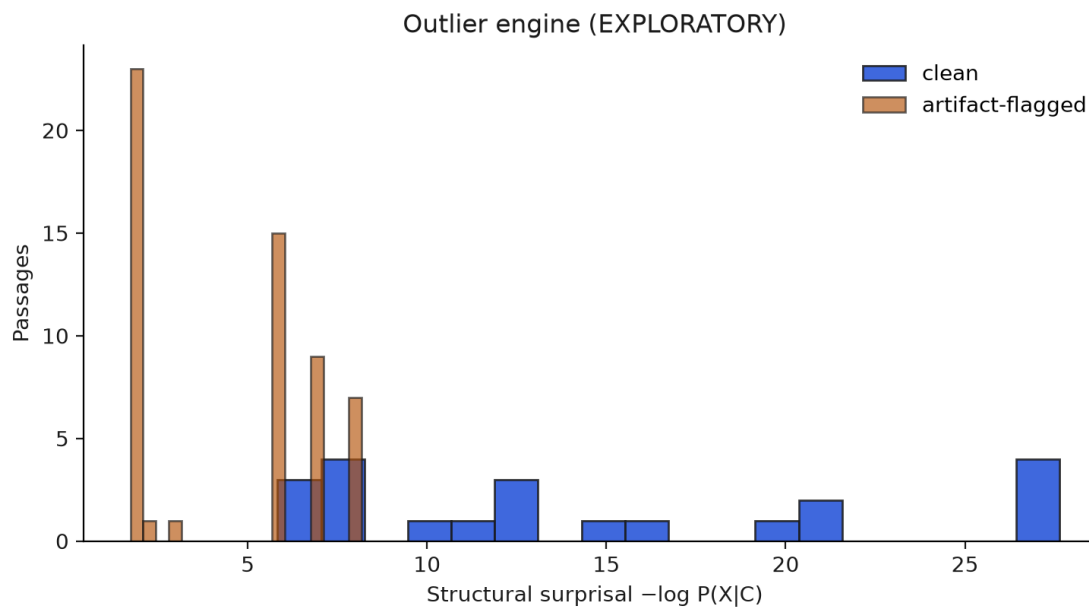


Figure 27: Structural surprisal with artifact flags.

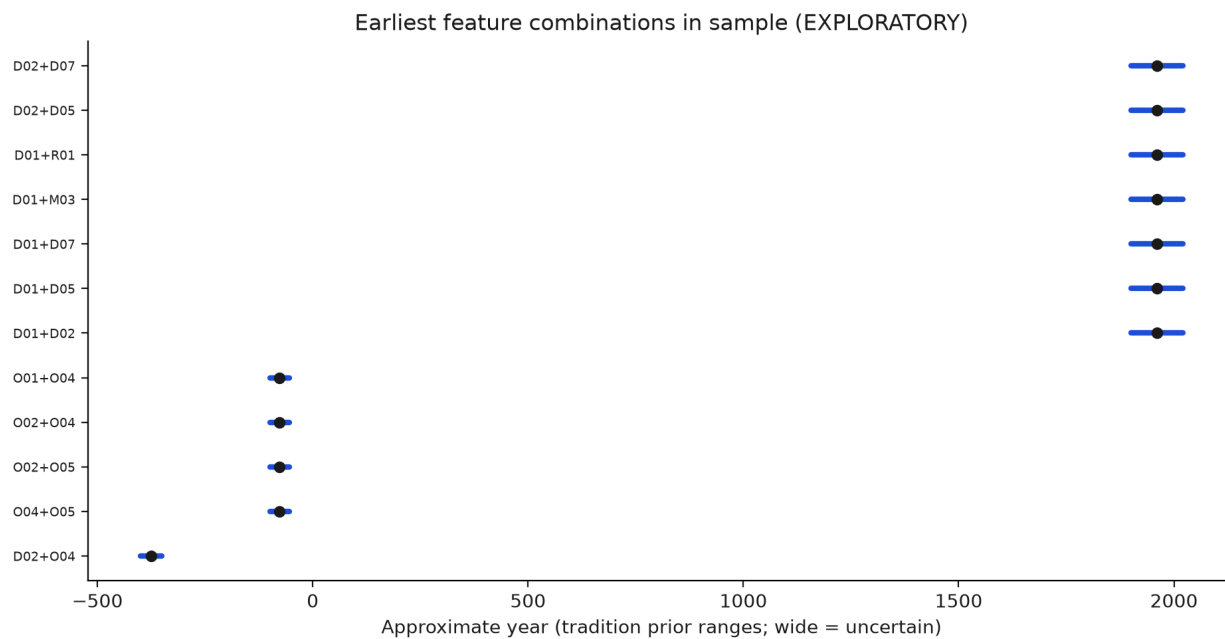


Figure 28: Earliest feature combinations using wide tradition date priors.

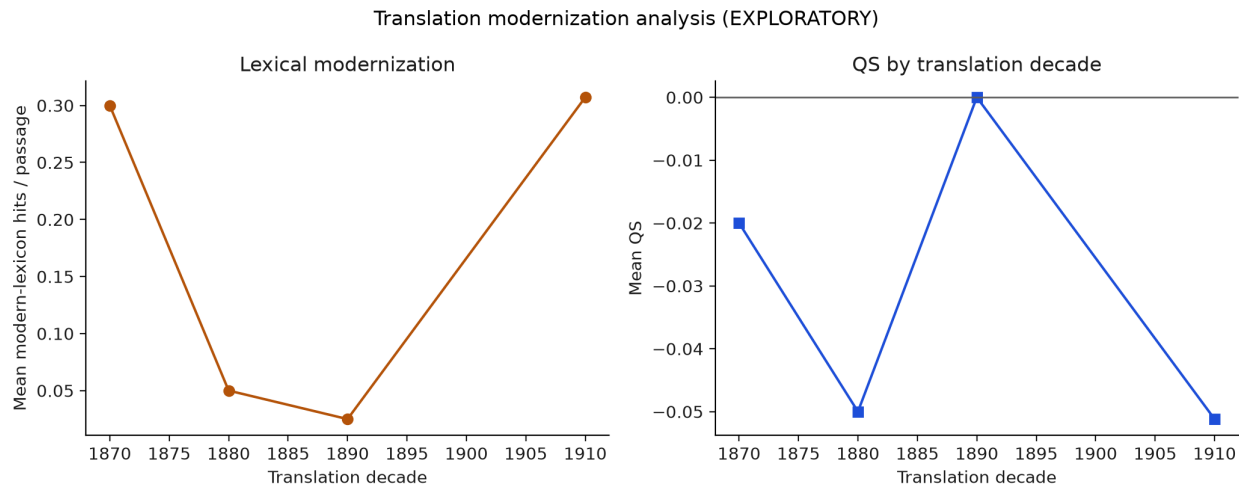


Figure 29: Lexical modernization vs QS by translation decade (exploratory).

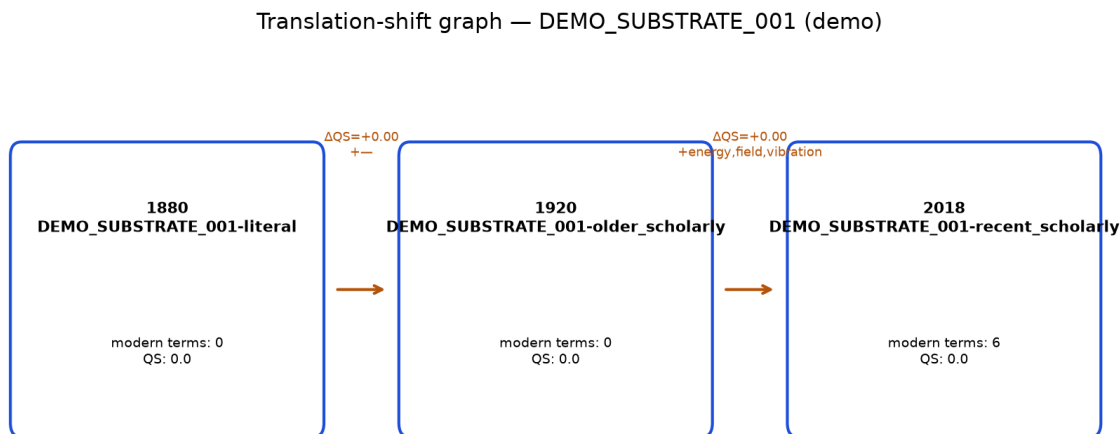


Figure 30: Aligned translation-shift graph (synthetic multi-translation demo).

Synthetic translation demo: injecting “quantum/energy/field/vibration” wording yielded TCI ≈ 0 (Figure 31).

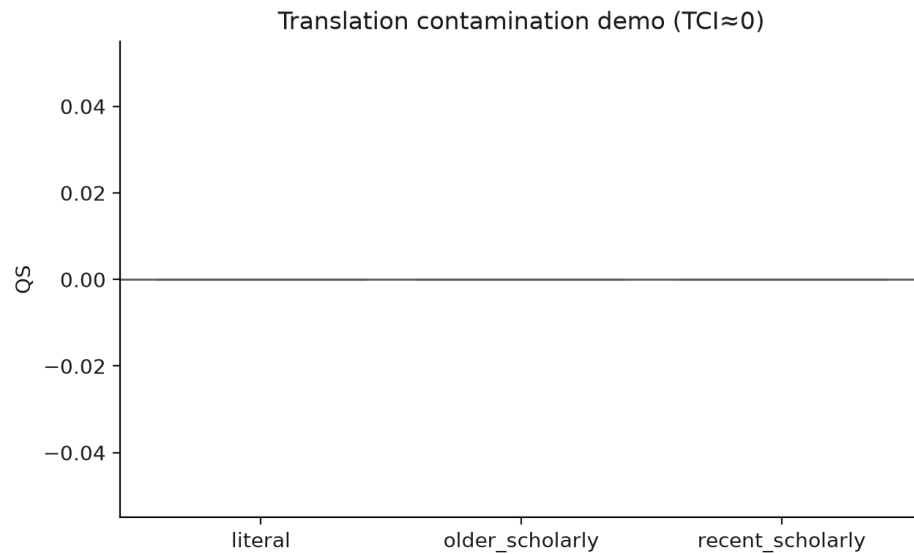


Figure 31: Synthetic translation-contamination demo.

13.3 Claims vs data

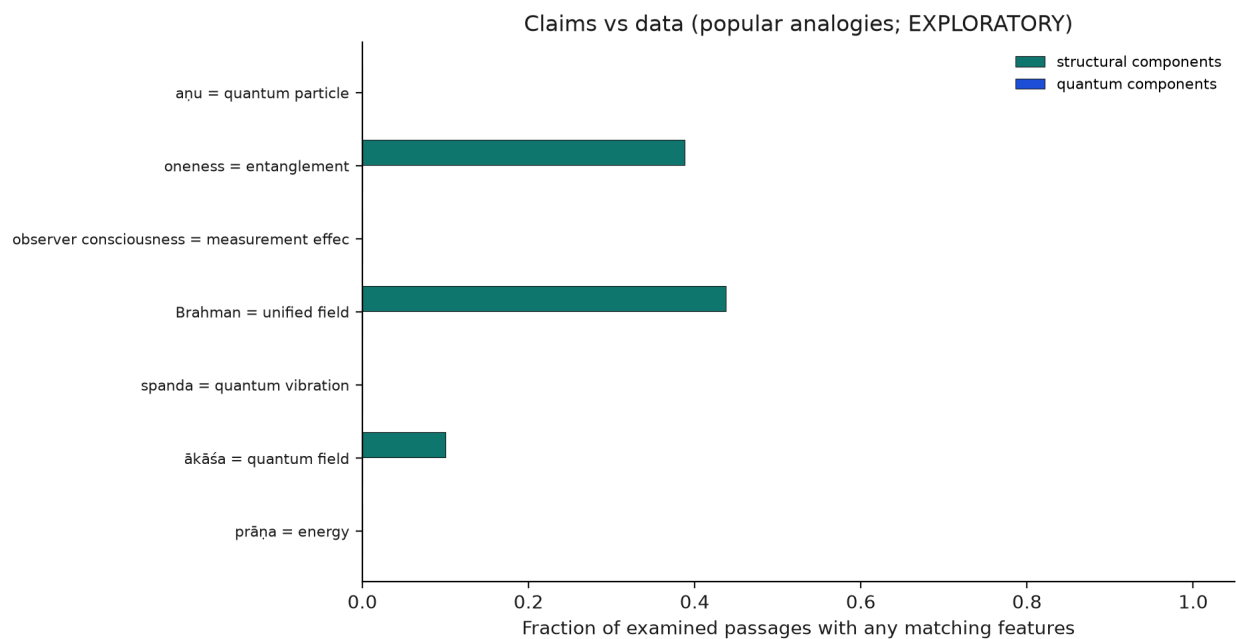


Figure 32: Popular claim components vs quantum components on Vedānta PD slice.

Popular quantum analogies are tested as structured claims. Field-like support without Level III features is reported as classical/field-like match—not as quantum confirmation and not as failure.

13.4 Novelty gating

No candidate was promoted to `STRONG_DISCOVERY_CANDIDATE`. Automated literature search leaves all top motifs at `NOVELTY_REVIEW_REQUIRED`. We do not claim “first ever.”

14 Robustness

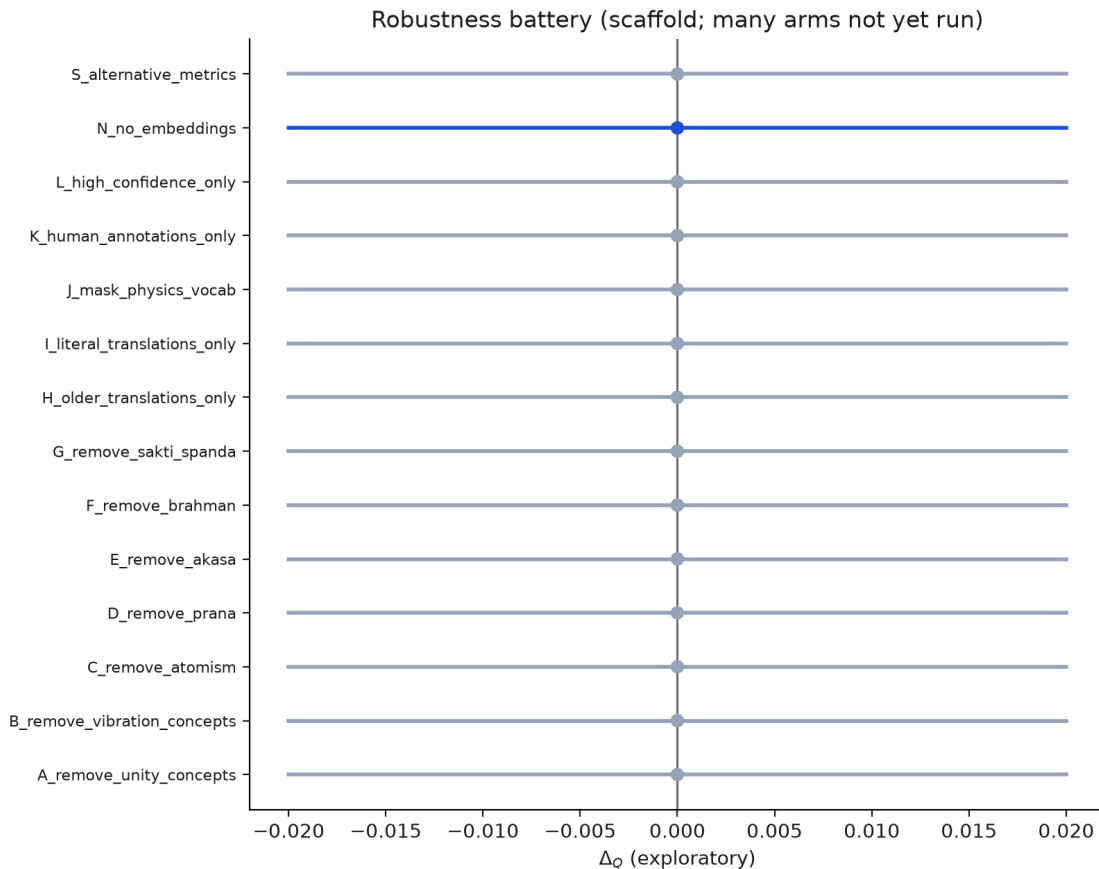


Figure 33: Robustness battery scaffold.

15 Discussion

15.1 What is established so far

1. The ontology distinguishes classical fields from quantized fields and unity from nonseparability on controls.
2. A dual confirmatory/discovery architecture is implemented with an explicit novelty gate.
3. Under heuristic PD annotation, System B finds field-like/compositional structure without quantum-specific motif enrichment for the Sanskrit target slice.

15.2 What remains

Freeze ontology after further piloting; licensed multi-translation pairs; human validation; expert review; OSF preregistration; LLM annotation; confirmatory unlock once.

15.3 Interpretation ladder

Textual similarity → structural analogy → statistical enrichment → quantum-vs-classical specificity → multiple independent Level III features. Even the top rung is not historical discovery of QM.

16 Limitations

Heuristic annotators are brittle; PD English is not a substitute for philologically controlled Sanskrit editions; dating uses wide priors; development decisions contaminate exploratory data; novelty search is automated and incomplete. See docs/LIMITATIONS.md.

17 Conclusion

RISHI-Q is both a confirmatory instrument and a discovery engine. The present manuscript freezes the methodological story, demonstrates instrument behavior on controls, and reports exploratory System B outputs and Capra-style claim autopsies without manufacturing significance or inventing ancient quantum discovery. Under heuristic public-domain annotation, Level I metaphysical structure appears without Level III quantum enrichment for the Vedānta slice examined here—a result that constrains popular rhetoric even while confirmatory H1 remains locked. Confirmatory evaluation awaits preregistration, human validation, and stronger annotation backends. Rigor first; discovery second; extraordinary claims only with extraordinary evidence.

Data and code availability

Software: MIT-licensed repository (<https://github.com/arjunkshah/rishi-q>). Private blinding maps are not published. Regenerate exhibits:

```
uv run python scripts/run_discovery_engine.py
uv run python scripts/run_capra_autopsy.py
uv run python scripts/build_all_visualizations.py
uv run python scripts/build_paper_assets.py
```

Acknowledgments

External Sanskrit, physics, and statistics reviewers are invited; none are invented here. No human subjects were enrolled for this textual study.

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- A Ontology feature inventory
- B Exploratory passage scores
- C Frequent positive features

Table 4: RISHI-Q ontology v0.1 feature inventory.

id	name	family	level	quantum_specific	field_like
O01	Fundamental substrate	fundamental_ontology	I	False	False
O02	Constituents	fundamental_ontology	II	False	False
O03	Fundamental discreteness	fundamental_ontology	II	False	False
O04	Continuous substrate	fundamental_ontology	II	False	True
O05	Local manifestation	fundamental_ontology	II	False	True
O06	Emergence	fundamental_ontology	II	False	False
D01	State	dynamics	II	False	False
D02	Transformation	dynamics	II	False	False
D03	Periodicity	dynamics	II	False	True
D04	Propagation	dynamics	II	False	True
D05	Conservation	dynamics	II	False	False
D06	Symmetry	dynamics	II	False	False
D07	Reversibility	dynamics	II	False	False
Q01	Discrete allowed states	quantum_specific	III	True	False
Q02	Superposed possibilities	quantum_specific	III	True	False
Q03	Fundamental probability	quantum_specific	III	True	False
Q04	Measurement-conditioned outcome	quantum_specific	III	True	False
Q05	Incompatible observables	quantum_specific	III	True	False
Q06	Nonseparability	quantum_specific	III	True	False
Q07	Context dependence	quantum_specific	III	True	False
Q08	Quantized excitation	quantum_specific	III	True	True
R01	Local causation	relational	II	False	True
R02	Long-range relation	relational	II	False	False
R03	Holistic state	relational	II	False	False
R04	Relational properties	relational	II	False	False
M01	Observer/system distinction	observer_epistemic	II	False	False
M02	Observation modifies system	observer_epistemic	III	True	False
M03	Epistemic limitation	observer_epistemic	I	False	False
M04	Ontic indeterminacy	observer_epistemic	III	True	False
F01	Spatial distribution	field_like	II	False	True
F02	Local state of distributed entity	field_like	II	False	True
F03	Disturbance propagation	field_like	II	False	True
F04	Localized manifestation	field_like	II	False	True
F05	Interaction mediation	field_like	II	False	True
F06	Excitation-like structure	field_like	II	False	True
F07	Quantization of distributed structure	field_like	III	True	True

Table 5: Exploratory passage scores (synthetic + physics controls). Not confirmatory.

passage_id	tradition	role	QS	QEF	field_class
PHYS_NEWTON_001	modern_physics	physics_reference	-0.500	0.000	insufficient
PHYS_EM_001	modern_physics	physics_reference	-0.100	0.000	classical_field_li
PHYS_THERMO_001	modern_physics	physics_reference	-1.000	0.000	insufficient
PHYS_QM_001	modern_physics	physics_reference	0.900	1.000	insufficient
PHYS_QFT_001	modern_physics	physics_reference	0.333	1.000	potentially_qft_l
PHYS_ENTANGLE_001	modern_physics	physics_reference	0.800	1.000	insufficient
SYN_UNITY_001	synthetic_metaphysical	negative_control	0.000	0.000	insufficient
SYN_FIELD_001	synthetic_target_like	target	0.000	0.000	metaphor_substr
SYN_ATOM_001	synthetic_vaisheshika_like	target	-0.333	0.000	insufficient
SYN_MYSTICAL_001	synthetic_mystical	control	0.000	0.000	insufficient
SYN_GREEK_001	greek	control	-0.333	0.000	insufficient

Table 6: Most frequent positive ontology features in the exploratory run.

feature_id	n_positive
O04	3
R01	3
D04	3
Q03	3
O05	2
O03	2
O02	2
M03	2
Q06	2
F05	2
F04	2
F03	2
F01	2
D05	2
Q05	1
Q04	1
Q07	1
Q02	1
Q01	1
Q08	1
D01	1
M04	1
O01	1
D02	1
M02	1